MAST30027: Modern Applied Statistics

Assignment 2, 2026 — Great Barrier Reef Consulting Report

AUTHOR
Student name and ID

Due: 5:00 pm Friday 11 September 2026

Late submissions will not be accepted unless an extension has been granted by the Special Consideration team.

This assignment is worth **12%** of your final mark. It has **27 marks**.

Submission and working

Create a report using the supplied QMD template and replace the author placeholder with your name and student ID. Create an RStudio Project for the assignment, keep the QMD and CSV files together in the project folder, and use relative file paths. If you include images of handwritten working, keep them in an `images` folder inside the same project.

Handwritten working

Mathematical working may be typed using LaTeX or completed clearly by hand and included as a picture in the relevant part of the report. If you use handwritten working, scan or photograph it, crop it closely, and save it as a PNG image. The image must be upright, high contrast, legible at normal PDF magnification, and labelled with the relevant part number.

Store the image in the `images` folder beside your QMD. Include it by placing `knitr::include_graphics()` **inside an R code chunk**, as illustrated below. Replace the example path with the relative path to your own image and remove `#| eval: false` when the file is present.

```
knitr::include_graphics("images/part4-working.png")
```

Handwritten working must appear in the rendered report and final PDF. It may support mathematical calculations, but it does not replace required R code, output, or written interpretation. Do not submit separate image files.

Final submission

Render the QMD to HTML, open the HTML in a browser, and use **Print** and **Save as PDF** to create `studentID_assignment2.pdf`. Check that all code, output, equations, tables, figures, and handwritten working are readable and are not cut off. Submit this single PDF through the LMS. Do not submit the QMD, HTML, CSV, or separate image files. It is your responsibility to upload the correct PDF on time.

Consulting engagement

> **Your role and client**
>
> You are a **Graduate Statistical Consultant** at **Meridian Insights**, a fictional analytics consultancy. A reef-monitoring program has asked your team to analyse Great Barrier Reef survey data and prepare advice about patterns in coral trout abundance.

You must submit a **consulting report**, not a sequence of disconnected answers. The report will be read by:

1. **Reef managers**, who want concise conclusions, uncertainty, and practical monitoring advice in plain language.
2. **Meridian Insights' technical reviewer**, who must be able to verify your GLM formulation, R code, model comparisons, diagnostics, and reasoning.

Your report should move coherently from the client's question to exploration, modelling, model checking, conclusions, and limitations. Explain why each analytical step is needed and what you learn from it. Code and output alone are not analysis.

Scientific background

Coral trout in the genus *Plectropomus* are conspicuous reef fishes and important fishery species on the Great Barrier Reef. The Australian Institute of Marine Science (AIMS) Long-Term Monitoring Program uses standardised underwater visual censuses along fixed transects to monitor reef fishes.

For further scientific context, see the AIMS pages on fish research and monitoring and the Long-Term Monitoring Program. These pages provide useful background about reef fishes and the monitoring program; conclusions in your report must be based on the supplied dataset and your own analysis.

This assignment examines whether the expected coral trout count at a monitoring site is associated with:

- the site's broad north–south latitude band; and
- whether the reef was open to fishing or in a no-take zone.

The analysis is observational. An association between zone and coral trout abundance should not automatically be interpreted as a causal effect of zoning.

Data source and preparation

The supplied file `aims_coral_trout_sites_2012.csv` is a teaching extract derived from the AIMS Long-Term Monitoring Program visual census fish dataset. The source was the AIMS NERP zoning-program file `nerp_ltmp_fish_zoning.csv`.

The data have been prepared for this task as follows:

1. Survey records from 2012 were selected, so the analysis concerns a single survey year.
2. Counts of all recorded *Plectropomus* species were added within each transect. A surveyed transect with no *Plectropomus* record was assigned a

count of zero.
3. The five transect counts at each site were added to form a **site-level total**.
4. Site identifiers were anonymised.
5. Sites were placed into three broad latitude bands using the recorded reef latitude:
    - `Northern`: latitude north of 17.9°S;
    - `Central`: latitude from 17.9°S to 19°S; and
    - `Southern`: latitude south of 19°S.

These latitude bands were created for this assignment and are not official AIMS management regions. The resulting file contains 120 site-level observations. Every row summarises five equal-length transects, so the site-level counts are directly comparable.

> **Data acknowledgement**
>
> Data were sourced from the Australian Institute of Marine Science (AIMS). Retain this acknowledgement and the source link in your submitted report.

Variables

Variable	Description
`site`	Anonymised monitoring-site identifier
`latitude_band`	`Northern`, `Central`, or `Southern`
`zone`	`Open` or `No-take`
`coral_trout`	Total number of coral trout recorded across five transects at the site

Keep the CSV file in the same folder as this QMD file. Load and prepare it using:

```
reef <- read.csv("aims_coral_trout_sites_2012.csv")

reef$latitude_band <- factor(
  reef$latitude_band,
  levels = c("Northern", "Central", "Southern")
)
reef$zone <- factor(
  reef$zone,
  levels = c("Open", "No-take")
)

dim(reef)
names(reef)
str(reef)
summary(reef)
```

Remove `#| eval: false` once the data file is correctly located and you are ready to begin your analysis.

Client question

The Monitoring Manager asks:

> "In the 2012 surveys, how did the expected coral trout count vary among the three latitude bands and between sites on reefs open to fishing and in no-take zones? Did the association with zone differ across latitude bands? How certain are the results, and what should we conclude—or avoid concluding—from this analysis?"

Required report

Write a professional consulting report that answers the client's question using only the GLM and overdispersion methods covered in the relevant lectures. Deciding on an appropriate initial model is part of the assignment.

Model choice is assessed

You must decide on an appropriate response distribution and link function from those covered in the subject. Justify both choices using the nature of the response and the model assumptions. In particular, explain how the chosen link ensures that fitted expected counts take valid values and how the model coefficients can be interpreted for the client.

The report contains nine assessed parts, each worth **3 marks**, for a total of **27 marks**.

Part 1: Executive summary (3 marks)

Provide a concise, non-technical summary of the problem, statistical approach, principal findings, uncertainty, and most important limitation. This section should stand on its own and should not contain R code.

Part 2: Problem and data understanding (3 marks)

Translate the client's questions into statistical objectives. Identify the response and explanatory variables, describe the observational unit and experimental design, and check the dataset's structure, factor levels, and missing values.

Part 3: Exploratory analysis (3 marks)

Use appropriate numerical summaries and plots to investigate the response distribution, differences across latitude bands and zones, and whether the association with zone appears to differ across latitude bands. Explain why each display is useful and interpret the important patterns. Figures must show the underlying observations or otherwise make the amount of data in each group clear.

Part 4: Initial GLM choice and formulation (3 marks)

Decide on an appropriate GLM for the response, including its response

distribution and link function. Explain the complete statistical model: define the random variable and its mean, state the distribution and mean–variance relationship, identify the assumptions being made about the observations, specify the link function, and write the linear predictor containing `latitude_band`, `zone`, and their interaction. Justify why the distribution and link are appropriate, including how the link ensures that fitted expected counts take valid values. Fit the model in R and explain the meanings of the intercept, main effects, and interaction under the chosen link. A `glm()` command and its output without this explanation are not sufficient.

Part 5: Initial model comparison (3 marks)

Compare the interaction model with the corresponding additive model. Use an appropriate difference-in-deviance procedure for your initial GLM and compare AIC values where valid. State which mean structure appears preferable at this stage and justify the decision. Do not interpret a main-effect coefficient in isolation if an interaction is retained.

Part 6: Dispersion assessment (3 marks)

Assess whether the variance assumption of your initial model is adequate. Calculate and interpret the estimated dispersion using Pearson residuals. Support your conclusion with an appropriate goodness-of-fit calculation and explain what the result implies for inference from the initial model.

Part 7: Overdispersion model and final selection (3 marks)

If there is evidence of overdispersion, fit an appropriate model covered in the overdispersion lectures and revisit the interaction using the correct comparison for nested models. Explain which model-selection tools are and are not valid for this revised model, how accounting for overdispersion changes inference, and which final model you select. If you find no evidence of overdispersion, justify retaining the initial variance assumption.

Part 8: Diagnostics and model alternatives (3 marks)

Assess the final model using diagnostic methods covered in the GLM lectures. Include:

- residuals plotted against the fitted linear predictor or fitted mean;
- an appropriate assessment of unusually large residuals;
- leverage; and
- Cook's distance or another influence measure covered in the subject.

Identify any site that appears unusual or influential, and explain what the diagnostics indicate. If warranted, refit the same model without that site and compare the substantive conclusions to investigate its influence. Discuss whether the diagnostic results suggest that an alternative model covered in the subject should be considered. Do not delete a site merely because it is inconvenient.

Part 9: Estimates and uncertainty for the client (3

marks)

Use the final model to answer the client's questions. Present fitted mean counts at site level for the relevant combinations of latitude band and zone. Calculate an estimate and 95% confidence interval for the comparison between sites in no take zones and sites on reefs open to fishing. Calculate the interval on the scale of the linear predictor and, where appropriate, transform the estimate and interval endpoints back to the response scale. If the interaction term is retained, interpret the relevant coefficients together. Use the results to answer the client's questions directly.

Presentation requirements

Each part is worth 3 marks. Marks may be lost for unsupported claims, unreadable displays, unexplained output, or conclusions that do not follow from the analysis.

- Create your report using this QMD template and write beneath the supplied headings. You may add subheadings, but do not delete the assignment brief or data acknowledgement.
- Replace the author placeholder in the YAML with your name and student ID.
- The report, from **Part 1** through **Part 9**, should be no more than **2,000 words**. Figure captions, table contents, equations, R code, and R output do not count toward this limit.
- Integrate all working into the relevant assessed part. The mathematics, R code, output, tables, figures, diagnostics, and interpretation supporting a conclusion must appear together in that part. No separate section for working is permitted.
- All R code and results used in the analysis must be visible. Do not paste screenshots of code, console output, tables, or figures.
- The QMD must render successfully from beginning to end in a clean R session and must not rely on objects created interactively.
- Include every package-loading command required by your report. Installation commands such as `install.packages()` must not appear in the submitted report.
- Use a two-sided significance level of 5% and 95% confidence intervals unless stated otherwise.
- Report numerical results to a sensible precision. Give every table and figure an informative title or caption, label axes clearly, and refer to each display in the text.
- Use the GLM and quasi-likelihood methods covered in the subject. Cite any external sources, packages, or code you use.

Part 1: Executive summary

Write this section for the Reef Monitoring Manager. Briefly state the problem, the modelling approach, the main findings and their uncertainty, and the most important limitation. It should stand alone, use non-technical language, and contain no R code.

Part 2: Problem and data understanding

State the statistical questions, response, explanatory variables, observational unit, and experimental design. Explain what one row and one count represent. Show concise checks of the dataset dimensions, variable types, factor levels, and missing values, and explain what these checks establish.

Part 3: Exploratory analysis

Include numerical summaries and clearly labelled plots that show the response distribution, the observations within each latitude-band-by-zone group, and any apparent interaction pattern. Explain why each display is useful and describe the features that matter for the subsequent modelling.

Part 4: Initial GLM choice and formulation

Describe the model before interpreting the R output. Define Y_i *and* μ_i*, state the response distribution, variance function, observation assumptions, and link, and write* $g(\mu_i) = \eta_i$ *with both main effects and their interaction. Explain why each component is suitable for these data, then show the* `glm()` *fit and interpret the coefficients under the chosen link.*

Part 5: Initial model comparison

Fit the corresponding additive model and compare it with the interaction model using the procedures requested above. Report the hypotheses, test result, and relevant model-comparison values. Make a provisional decision about the mean structure and explain it in context.

Part 6: Dispersion assessment

Calculate the Pearson dispersion estimate and an appropriate goodness-of-fit result for the initial model. Show the calculation, interpret both quantities, and explain what they imply about the variance assumption and the reliability of the initial standard errors and tests.

Part 7: Overdispersion model and final selection

If required by Part 6, refit the relevant mean structures using an overdispersion method covered in class and compare the nested models correctly. Explain how the revised model changes uncertainty and which selection procedures are valid. State and justify your final model.

Part 8: Diagnostics and model alternatives

Present the requested residual and influence diagnostics for the final model, with

readable labels or a clear way to identify relevant sites. Describe any patterns or unusual observations. If warranted, refit the same model without an influential site and compare the main conclusions. Discuss whether the results suggest an alternative model covered in the subject, and do not remove an observation from the primary analysis without justification.

Part 9: Estimates and uncertainty for the client

Present the fitted mean counts at site level for the relevant combinations of latitude band and zone. Calculate a 95% confidence interval for the comparison between sites in no take zones and sites on reefs open to fishing. Calculate the interval on the scale of the linear predictor and, where appropriate, transform the estimate and interval endpoints back to the response scale. If the interaction is retained, interpret the relevant coefficients together. Use these results to answer the client's questions in context.